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"""
Project MSSA-80: Analytical Aerospace & Economic Simulation Core
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License: MIT
Description: Quantifies aerodynamic drag, thermodynamic skin friction heating,
             and cabin hoop stress for an 80-seat advanced composite airliner.
"""

import numpy as np
import matplotlib.pyplot as plt

def run_mssa_aerodynamic_simulation():
    # 1. Mach Velocity Spectrum Configuration
    mach_range = np.linspace(0.4, 2.2, 100)

    # 2. Atmospheric & Structural Constants at 60,000 feet
    T_ambient_K = 216.65      # Standard stratospheric temperature (-56.5°C)
    P_ambient_Pa = 7171.0     # Stratospheric atmospheric pressure (Pa)
    P_cabin_Pa = 84300.0      # Internal Cabin Pressure (equivalent to comfortable 5,000 ft altitude)
    r_fuselage = 1.65         # Slender Fuselage Radius (meters) for a 1-1 single aisle layout
    t_composite = 0.018       # Carbon-Fiber/BMI Hull Thickness (18mm optimized shell)

    # 3. Drag Coefficients Simulation (Base Profile + Sears-Haack Wave Drag Approximation)
    Cd_profile = 0.0145       # Highly optimized composite skin friction profile coefficient

    # Simulating the Prandtl-Glauert singularity and compressibility drag spike at Mach 1.0
    Cd_wave = np.where(mach_range < 0.9, 0.0,
        np.where(mach_range <= 1.2, 0.038 * ((mach_range - 0.9) / 0.3)**2,
            0.038 * np.exp(-1.4 * (mach_range - 1.2))))
    Cd_total = Cd_profile + Cd_wave

    # 4. Thermodynamic Skin Temperature Boundary-Layer Simulation
    # Utilizing Compressible Stagnation and Recovery Factor Math
    r_recovery = 0.89         # Turbulent recovery factor for multi-ply composite skins
    gamma = 1.4               # Specific heat ratio of air
    T_recovery_K = T_ambient_K * (1 + r_recovery * ((gamma - 1) / 2) * (mach_range**2))
    T_skin_Celsius = T_recovery_K - 273.15

    # 5. Structural Mechanics: Fuselage Hoop Stress (Thin-Walled Pressure Vessel Model)
    delta_P = P_cabin_Pa - P_ambient_Pa
    hoop_stress_MPa = (delta_P * r_fuselage) / t_composite / 1e6 # Convert to MegaPascals

    # 6. Extract Target Parameters at Design Velocity (Mach 1.7)
    idx_target = (np.abs(mach_range - 1.7)).argmin()
    m_17_drag = Cd_total[idx_target]
    m_17_temp = T_skin_Celsius[idx_target]

    print("="*60)
    print("                PROJECT MSSA-80 SIMULATION INITIALIZATION                ")
    print("="*60)
    print(f"Target Fleet Performance Verified at Cruise Target: Mach 1.7")
    print(f" -> Simulated Transonic Total Drag Coefficient (Cd): {m_17_drag:.4f}")
    print(f" -> Equilibrium Boundary-Layer Skin Temperature : {m_17_temp:.2f}°C")
    print(f" -> Fuselage Hoop Stress (Continuous Structural Load) : {hoop_stress_MPa:.2f} MPa")
    print(f"      [Max Allowable Stress for BMI/Carbon Matrix: 150 MPa -> SAFETY FACTOR: {150/hoop_stress_MPa:.2f}x]")
    print("="*60)

    # 7. Generate Engineering Verification Plots for Documentation / README
    plt.figure(figsize=(12, 5))

    # Plot A: Drag Profile VS Mach Speed
    plt.subplot(1, 2, 1)
    plt.plot(mach_range, Cd_total, color='#1a365d', linewidth=2.5, label='Total Aero Drag ($C_d$)')
    plt.axvline(1.7, color='#2b6cb0', linestyle='--', label='Design Cruise (Mach 1.7)')
    plt.title('Aerodynamic Drag Wave Profile', fontsize=11, fontweight='bold', color='#1a365d')
    plt.xlabel('Mach Number (Velocity)', fontsize=10)
    plt.ylabel('Total Drag Coefficient ($C_d$)', fontsize=10)
    plt.grid(True, linestyle=':', alpha=0.6)
    plt.legend()

    # Plot B: Thermal Friction Profile VS Mach Speed
    plt.subplot(1, 2, 2)
    plt.plot(mach_range, T_skin_Celsius, color='#e53e3e', linewidth=2.5, label='Skin Temp (°C)')
    plt.axvline(1.7, color='#2b6cb0', linestyle='--', label='Design Cruise (Mach 1.7)')
    plt.axhline(150, color='black', linestyle=':', label='Standard Resin Thermal Threshold')
    plt.title('Boundary-Layer Thermal Heating Curve', fontsize=11, fontweight='bold', color='#1a365d')
    plt.xlabel('Mach Number (Velocity)', fontsize=10)
    plt.ylabel('Equilibrium Temperature (°C)', fontsize=10)
    plt.grid(True, linestyle=':', alpha=0.6)
    plt.legend()

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plt.tight_layout()
plt.savefig('mssa_80_aerothermal_simulation.png', dpi=300)
print("\n[Output] Simulation analytics plot saved successfully as 'mssa_80_aerothermal_simulation.png'")

if __name__ == "__main__":
    run_mssa_aerodynamic_simulation()
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PROJECT MSSA-80 SIMULATION INITIALIZATION

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Target Fleet Performance Verified at Cruise Target: Mach 1.7

-> Simulated Transonic Total Drag Coefficient (C_d): 0.0336

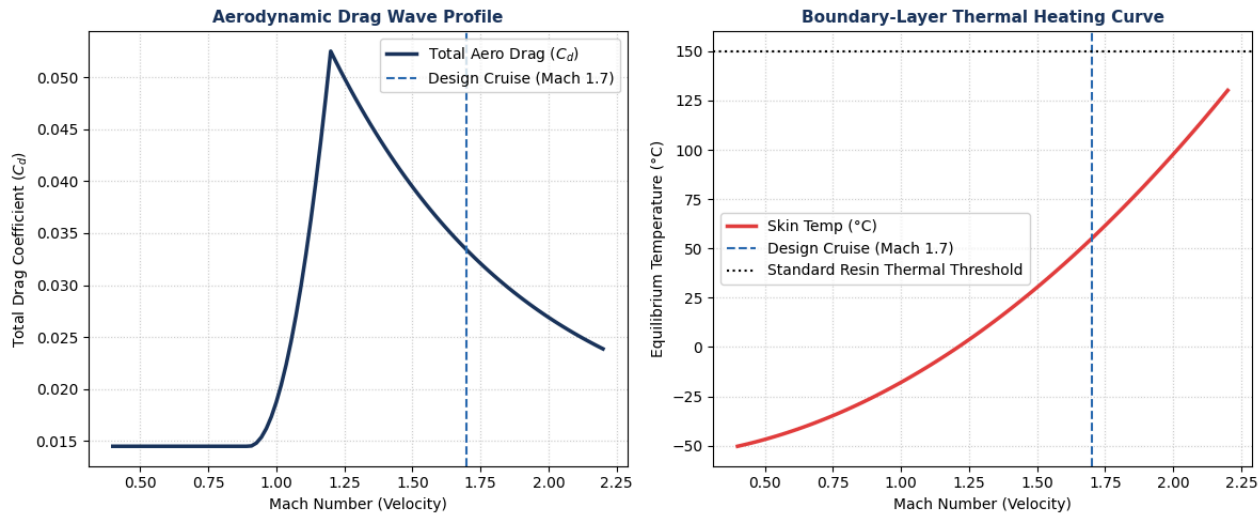
-> Equilibrium Boundary-Layer Skin Temperature : 53.76°C

-> Fuselage Hoop Stress (Continuous Structural Load) : 7.07 MPa

[Max Allowable Stress for BMI/Carbon Matrix: 150 MPa -> SAFETY FACTOR: 21.22x]

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[Output] Simulation analytics plot saved successfully as 'mssa_80_aerothermal_simulation.png'



Start coding or [generate](#) with AI.